This analysis examines whether transcriptomic differences across organ-tropic metastatic sublines reflect adaptation to distinct microenvironments. Results are presented for dimensionality reduction and differential expression analyses.

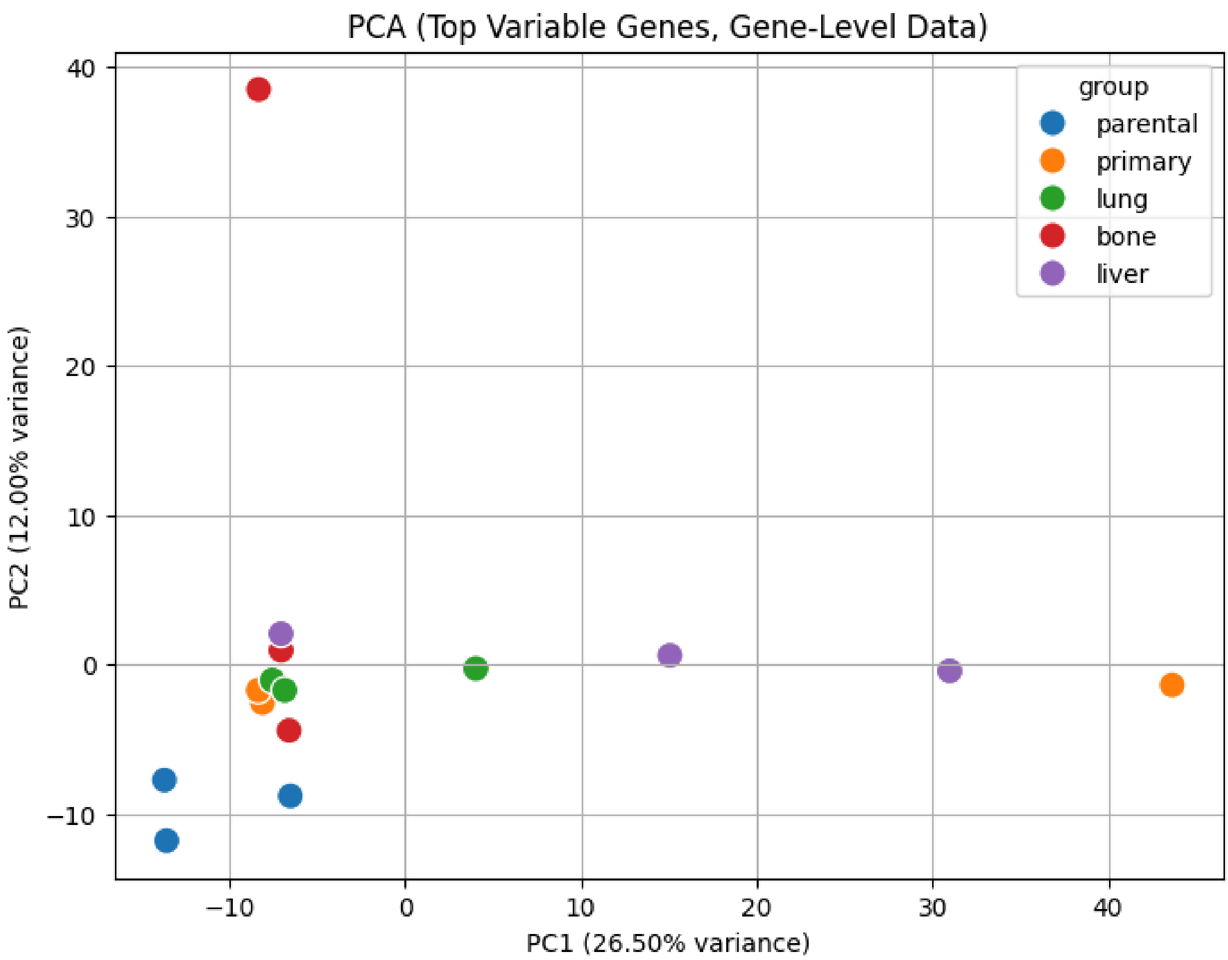

Figure 1: PCA of gene-level expression (top 1000 most variable genes) showing separation of metastatic sublines. PC1 (26.5% variance) captures a dominant axis along which liver-tropic samples are distributed, indicating a directional transcriptional shift rather than tight clustering. Lung samples cluster closely near the origin, suggesting a more stable transcriptional state. Bone samples show pronounced heterogeneity, including a strong outlier along PC2 (~38), indicating multiple transcriptional strategies associated with bone metastasis. Primary tumors also exhibit variability, with one extreme sample along PC1, suggesting pre-existing transcriptional diversity. This pattern suggests that liver metastasis may require a more coordinated transcriptional shift, whereas bone metastasis may tolerate multiple transcriptional states.

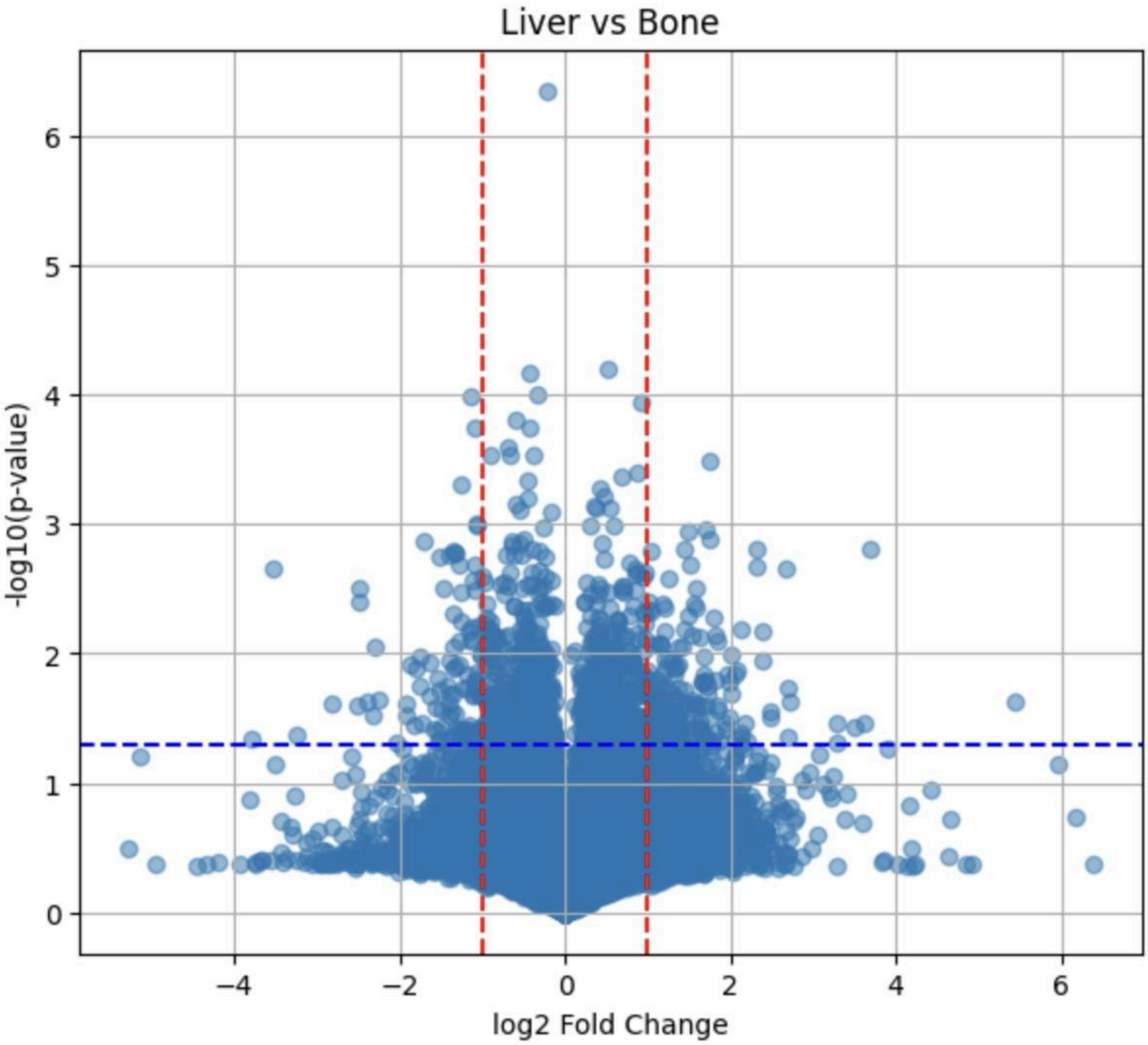

Figure 2: Volcano plot comparing liver- and bone-tropic metastatic sublines. A broad distribution of log2 fold changes (approximately -5 to +6) indicates substantial transcriptional divergence between the two groups. A larger number of genes exceed fold-change thresholds compared to liver vs lung, consistent with greater biological separation. While no genes meet strict FDR significance due to limited sample size, the presence of numerous moderately shifted genes suggests distributed transcriptional differences rather than a small number of dominant drivers. This suggests that bone metastatic adaptation may involve more extensive transcriptional reprogramming compared to liver, potentially reflecting differences in microenvironmental constraints.

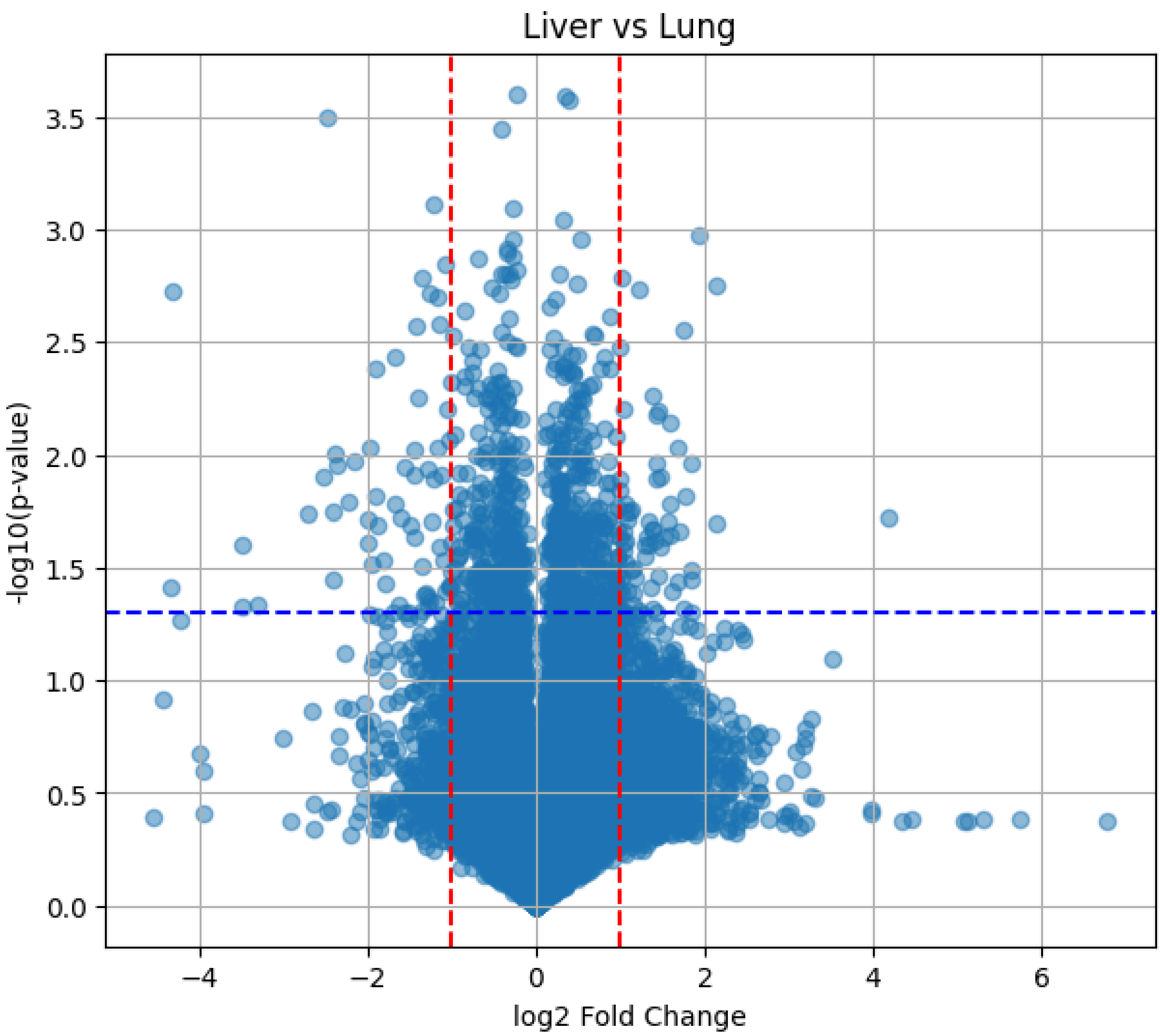

Figure 3: Volcano plot comparing liver- and lung-tropic metastatic sublines. Most genes cluster tightly around log2 fold change ≈ 0, indicating relatively modest transcriptional differences between these groups. The narrower spread compared to liver vs bone suggests a more similar transcriptional landscape. Few genes exceed fold-change thresholds, supporting the observation that liver and lung metastases share overlapping gene expression programs, with differences likely arising from subtle or pathway-level regulation. Despite modest gene-level differences, the PCA results suggest that these similarities may mask coordinated shifts at the pathway level rather than large individual gene changes.

Together, these results support a model in which organ-specific metastatic fitness arises from context-dependent transcriptional adaptation, with liver and lung metastases sharing overlapping programs and bone metastasis exhibiting greater heterogeneity and divergence.